Question

This question does not require stereochemistry. Figure 1 shows a two-step reaction.

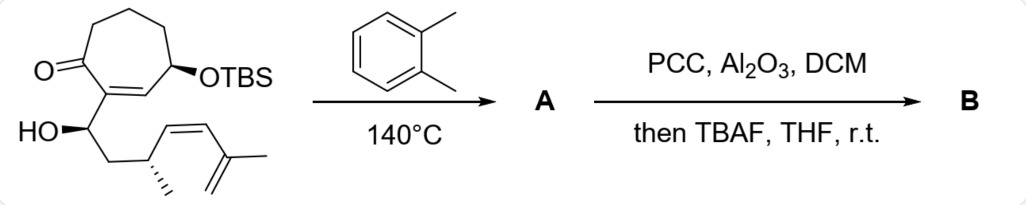

Fig. 1, the figure shows a two-step sequential reaction. The first step, described by SMILES, is: C=C(C)/C=C\[C@H](C)C[C@H](C1=C[C@@H](CCCC1=O)O[Si](C)(C)C(C)(C)C)O>[A], reaction conditions: o-xylene, 140℃. The second step, described by SMILES, is: [A]>>[B], reaction conditions: PCC (pyridinium chlorochromate), Al_2O_3 (aluminum oxide), DCM (dichloromethane), then TBAF (tetrabutylammonium fluoride), THF (tetrahydrofuran), room temperature.

Deduce the structural formula of $\mathbf{A}$.

In the subsequent transformation of $\mathbf{A}$, an unexpected reaction occurred, yielding compound $\mathbf{B}$. It is known that $\mathbf{B}$ contains only 3 rings, and $\mathbf{B}$ does not contain a hydroxyl group. Deduce the structural formula of $\mathbf{B}$ and the structural formula of the key intermediate in the process from $\mathbf{A}$ to $\mathbf{B}$.

There are the following statements:

1. $\mathbf{A}$ does not contain a five-membered ring.

2. $\mathbf{A}$ contains four rings of seven members or less.

3. A carbon-carbon single bond is formed in the process of generating $\mathbf{B}$.

4. $\mathbf{B}$ is an α,β-unsaturated ketone.

Among the following options, the one with all correct statements and the most correct statements is:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 2,3,4

O. 1,2,3,4

Answer

Correct Answer: A

Detailed Explanation

The first step, high temperature, is the classical D-A reaction condition. The molecule contains exactly one dienophile and one diene, and the two undergo a D-A reaction to form compound **A**, as shown in Figure 2. Statement 1 is incorrect, and statement 2 is incorrect.

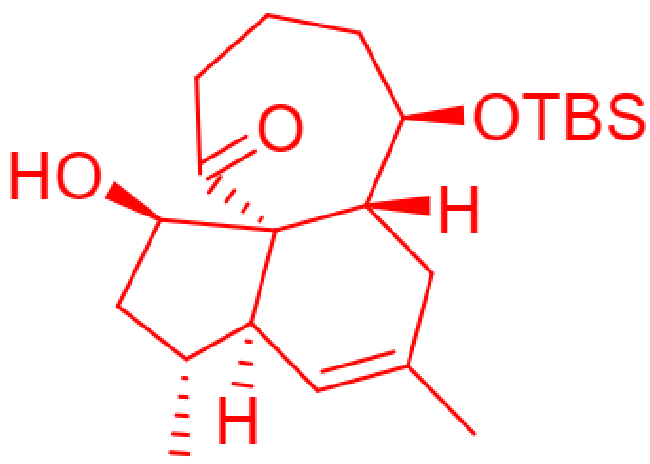

Fig. 2，The molecule in the figure is described by SMILES as:CC1=C[C@@]2(C)[C@H](C)C[C@H]([C@]32C(=O)CCC[C@H]([C@]3(C)C1)O[Si](C)(C)C(C)(C)C)O

CHECKPOINT 1 PTS

Intramolecular D-A reaction occurs at high temperature to generate the tricyclic product **A**

CHECKPOINT 1 PTS

A is described by SMILES as:CC1=C[C@@]2(C)[C@H](C)C[C@H]([C@]32C(=O)CCC[C@H]([C@]3(C)C1)O[Si](C)(C)C(C)(C)C)O

In the second step, PCC is a classical hydroxyl oxidizing agent, oxidizing the unprotected hydroxyl group in **A** to a carbonyl group. TBAF deprotects the silicon-containing protecting group, forming a hydroxyl anion. Generally, the product can be obtained after workup. However, the question mentions that an unexpected reaction occurred, and the product **B** does not contain a hydroxyl group, indicating that the hydroxyl group underwent further reaction. Consider possible reactions: The carbanion generated by the retro-aldol condensation/Michael addition reaction is unstable, which is excluded. It is found that the hydroxyl anion can attack the carbonyl group on the seven-membered ring, forming a stable five-membered and six-membered bridged ring structure. The structure of this intermediate is shown in Figure 3.

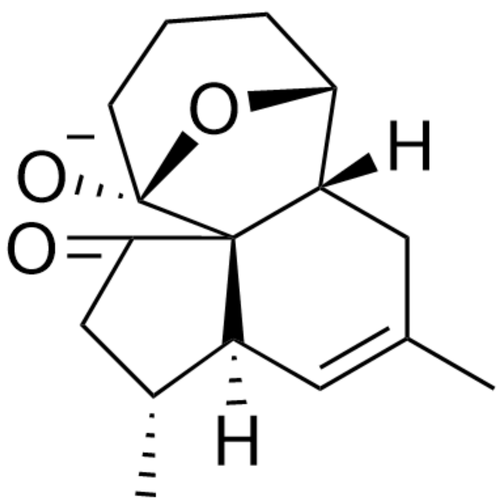

Fig. 3，The molecule in the figure is represented by SMILES as:CC(C[C@]1([H])[C@@]23[C@]4([O-])CCC[C@H]1O4)=C[C@@]3([H])[C@H](C)CC2=O

CHECKPOINT 1 PTS

After removing the protecting group, the hydroxyl group attacks the carbonyl group on the seven-membered ring to form a bridged ring intermediate.

CHECKPOINT 1 PTS

The structure of the intermediate is described by SMILES as:CC(C[C@]1([H])[C@@]23[C@]4([O-])CCC[C@H]1O4)=C[C@@]3([H])[C@H](C)CC2=O

At the same time, considering that there are only three rings in **B**, it indicates that carbon-carbon bond cleavage further occurred, reducing the number of rings. The hydroxyl anion formed by the attacked carbonyl group can further undergo an elimination reaction to generate a stable enolate anion, and product **B** is obtained after workup, the structure of which is shown in Figure 4.

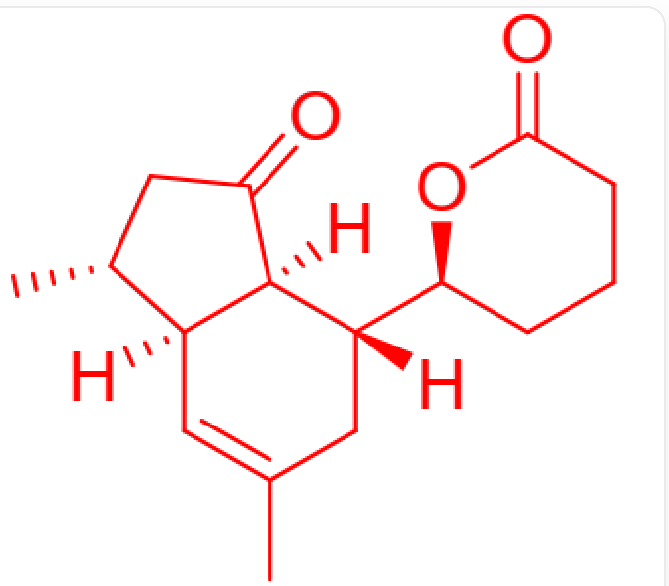

Fig. 4，CC1=C[C@]2([C@@H](CC([C@]2([C@](C1)([C@@H]3CCCC(O3)=O)[H])[H])=O)C)[H]

CHECKPOINT 1 PTS

A retro-aldol condensation reaction occurs to generate a stable enolate anion, and product $\mathbf{B}$ is obtained after workup

CHECKPOINT 1 PTS

Product $\mathbf{B}$, described by SMILES as:CC1=C[C@]2([C@@H](CC([C@]2([C@](C1)([C@@H]3CCCC(O3)=O)[H])[H])=O)C)[H]

In the process of generating $\mathbf{B}$, a carbon-oxygen double bond and a carbon-oxygen single bond are formed, and a carbon-carbon single bond is broken, but no carbon-carbon single bond is formed, so statement 3 is incorrect. $\mathbf{B}$ does not contain conjugated double bonds, so statement 4 is incorrect. Therefore, all the above statements are incorrect, and the answer is A